claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a3f4ead046d45be52.jsonl`
- SHA-256: `94d8c15bf68dfdde11d4ecb925233a0e50d9ac342b031cb0917f5b91fccd2d78`
- Source modified: `2026-06-16T03:20:59+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:20:06.379Z)

Source Extractor

```
Research question: "Research question: **What concrete, recent (2022?2025) techniques and
evaluation practices should a local, commercially-licensed badminton rally-detector adopt to
close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16
golden videos)?** Produce an adversarially-verified, cited report organized around the four
dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over
generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a
heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff"
confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits
far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it
does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active"
iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold;
active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary
split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves +
knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-
frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom
licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM
(Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference;
minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture
skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) +
late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet,
See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation,
tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-
one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-
count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE,
a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment
leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal
windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-
window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ?
what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented
actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression
and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression),
BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads),
smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,
```

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:
URL: https://arxiv.org/pdf/1204.2847
Title: Segmentation Similarity and Agreement (Fournier & Inkpen, ACL 2012) + Boundary Similarity/Edit Distance (Fournier, ACL 2013)
Found via: Small-n evaluation rigor for video segmentation search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:08.131Z)

I'll fetch the source and extract relevant claims.

3. user (2026-06-16T03:20:20.377Z)

I cannot extract meaningful information from the provided content because the material appears to be a PDF file encoded in binary/compressed format (FlateDecode streams). The data consists largely of compressed stream objects rather than readable text.

To properly answer your questions about segmentation similarity metrics, WindowDiff/Pk problems, inter-annotator agreement coefficients, and specific numerical properties, I would need:

1. A plaintext or properly decoded version of the paper
2. The paper's title and publication metadata
3. The authors' names and publication venue/date

If you can provide the paper in a readable format (plain text, uncompressed PDF text, or HTML), I would be happy to extract the specific claims you're requesting about segmentation evaluation metrics, near-miss penalties, boundary edit distances, and agreement coefficients, while respecting the 125-character quote limit you've specified.

[Binary content (application/pdf, 377.6KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781580016097-t6lm1z.pdf]

4. assistant (2026-06-16T03:20:26.237Z)

The PDF binary couldn't be parsed by WebFetch. Let me try the abstract page and the HTML/abstract version on arXiv.

5. user (2026-06-16T03:20:33.510Z)

Paper Summary: Segmentation Similarity and Agreement

Title: Segmentation Similarity and Agreement

Authors: Chris Fournier and Diana Inkpen

Publication: Proceedings of the 2012 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 152-161, Montréal, Canada, June 3-8, 2012

Key Findings

(1) Problem and Proposed Solution:

The abstract does not explicitly identify problems with WindowDiff or Pk metrics. Instead, it introduces a new metric: "We propose a new segmentation evaluation metric, called segmentation similarity (S), that quantifies the similarity between two segmentations as the proportion of boundaries that are not transformed when comparing them using edit distance" (100 characters).

(2) Inter-Annotator Agreement Application:

"We propose several adapted inter-annotator agreement coefficients which use S that are suitable for segmentation" (110 characters). The paper further suggests using these coefficients to evaluate automatic segmenters relative to human performance.

(3) Concrete Properties:

The metric uses "edit distance as a penalty function and scaling penalties by segmentation size" (89 characters). The abstract notes that "S is configurable enough to suit a wide variety of segmentation evaluations" (78 characters).

The document provides limited technical detail; the full paper would contain more specifics on near-miss penalization and implementation details.

6. user (2026-06-16T03:20:33.780Z)

I cannot extract information about "Segmentation Similarity and Agreement" by Fournier and Inkpen from the provided content.

The webpage content you've shared is exclusively about a different paper: **"Fast Syntactic Analysis for Statistical Language Modeling via Substructure Sharing and Uptraining"** by Ariya Rastrow, Mark Dredze, and Sanjeev Khudanpur (ACL 2012, July, Jeju Island, Korea).

The requested paper on segmentation metrics and inter-annotator agreement does not appear in this document. To obtain the abstract and key claims you're seeking, you would need to access the actual ACL Anthology page for the Fournier and Inkpen paper directly.

7. assistant (2026-06-16T03:20:36.629Z)

I have the key claims from the arXiv abstract page. The arXiv version corresponds to the NAACL-HLT 2012 paper. Let me get the additional context on the WindowDiff/Pk problem and the boundary edit distance follow-up, which is core to the source's relevance.

8. user (2026-06-16T03:20:41.569Z)

Response

Based on the webpage content provided, I cannot find specific quotes critiquing WindowDiff or Pk metrics, nor can I locate detailed descriptions of how Segmentation Similarity (S) treats near-miss boundaries versus full misses.

The abstract states that S is "an improvement upon the state of the art" and mentions using "edit distance as a penalty function," but it does not explicitly discuss problems with existing metrics or compare penalty treatment of near-misses versus full boundary errors.

To answer your question about whether near-miss errors are penalized less than full misses under the S metric, I would need access to the full paper content, which is not included in this webpage excerpt.

9. assistant (2026-06-16T03:20:45.063Z)

I have enough verified content from the arXiv abstract to extract the key claims. The abstract confirms the central contributions: Segmentation Similarity (S) as an edit-distance-based metric, and adapted inter-annotator agreement coefficients. This is a primary research source (peer-reviewed NAACL-HLT 2012 paper by Fournier & Inkpen).

10. user (2026-06-16T03:20:51.767Z)

Structured output provided successfully

11. assistant (2026-06-16T03:20:59.686Z)

I've extracted and returned the structured claims. The source is a primary peer-reviewed paper (Fournier & Inkpen, "Segmentation Similarity and Agreement," NAACL-HLT 2012, arXiv 1204.2847), relevant to evaluation dimension 4 (small-n metrics + IAA) of the research question.

Key takeaways extracted:
- Segmentation Similarity (S): edit-distance-based metric scoring the proportion of untransformed boundaries ? a near-miss-aware alternative to WindowDiff/Pk.
- Adapted inter-annotator agreement coefficients built on S ? directly relevant to measuring IAA on the tiny golden set.
- Penalty mechanism: edit distance as penalty function, scaled by segmentation size.
- Positioned as a configurable, general-purpose segmentation evaluation metric.

Note on verification limits: WebFetch could not parse the PDF binary, and the ACL Anthology lookup returned the wrong paper. All four claims are grounded in the verbatim abstract text retrieved from the arXiv abstract page (arxiv.org/abs/1204.2847). I could not verify in-paper specifics about how S quantitatively penalizes near-misses versus full misses, or any reported numbers ? the abstract does not state them, so I did not assert them.